



MAX-PLANCK-GESELLSCHAFT

Local entropy for finite time dynamics

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May 2019

Content

- Lagrangian coherent structures
- Asymptotic dynamics vs. Finite time dynamics
- Finite time Lyapunov exponents and Lagrangian coherent structures
- Finite time entropy (FTE)
- Applications to detection of invariant manifolds and coherent sets

Lagrangian coherent structures

Lagrangian coherent structures (LCSs) are distinguished surfaces of trajectories in a dynamical system that exert a major influence on nearby trajectories over a time interval of interest.

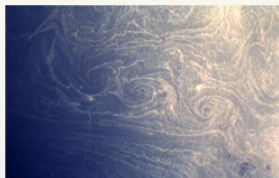
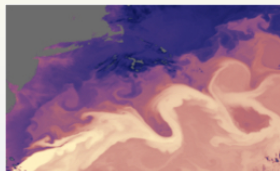


Figure: Lagrange coherent structures

Asymptotic dynamics vs finite time dynamics

- Systems is aperiodic and only defined in finite time intervals
- Many phenomena only happen in short time: vortex merger, tornados
- Classical concepts need to be redefined: stability, stable and unstable manifolds

Finite time dynamics - the setting

$$\dot{x} = f(t, x), \quad t \in I := [t_0, t_0 + T], x \in X \subset \mathbb{R}^n \quad (0.1)$$

where $T > 0$ and $f : I \times \mathbb{R}^d \rightarrow \mathbb{R}^d$, $f(\cdot, x) \in C^0(I)$, $f(t, \cdot) \in C^1(X)$.

$\varphi(\cdot, t_0, x_0)$ - solution of (0.1) starting at x_0 in t_0 .

$\Phi(\cdot, \cdot, x_0) \in \mathbb{R}^{n \times n}$ - the linearization of the solution $\varphi(t, t_0, x_0)$ of (0.1) w.r.t. x_0 , i.e. for $t \in I$, $\xi \in \mathbb{R}^d$: $\frac{\partial \varphi}{\partial x}(t, t_0, x) \xi = \Phi(t, t_0, x) \xi$.

Finite time Lyapunov exponents (FTLE)

$$\lambda_1(T, t_0, x_0) \geq \dots \geq \lambda_n(T, t_0, x_0)$$

$$\lambda_k(T, t_0, x) := \frac{1}{2T} \log \Lambda_k^2(\Phi(t_0 + T, t_0, x_0)^T \Phi(t_0 + T, t_0, x_0)), \quad k = 1, \dots, n$$

where $\Lambda_1^2(\Psi^T \Psi) \geq \dots \geq \Lambda_n^2(\Psi^T \Psi) \geq 0$ denote the eigenvalues of a nonnegative definite symmetric matrix $\Psi^T \Psi$.

$I_0 = \{t_0, t_0 + T\}$. Invariant differentiable manifold $M = (M(s))_{s \in I_0}$ of dimension $n - 1$, i.e. $\varphi(t_0 + T, t_0)M(t_0) = M(t_0 + T)$.

Tangent space $\Phi_{x_0}(t_0 + T, t_0)T_{x_0}M(t_0) = T_{\varphi(t_0+T, t_0)x_0}M(t_0 + T)$.

Normal direction: $\langle e_0, n_0 \rangle = 0, \forall e_0 \in T_{x_0}M(t_0)$.

- The *stretching rate* at $x_0 \in M(t_0)$:

$$\begin{aligned} r(x_0) &:= \frac{1}{T} \log \max_{v \in T_{x_0}M(t_0)} \frac{\|\Phi_{x_0}(t_0 + T, t_0)v\|}{\|v\|} \\ &= \frac{1}{T} \log \|\Phi_{x_0}(t_0 + T, t_0)\|_{T_{x_0}M(t_0)}. \end{aligned}$$

- The *normal repulsion rate*:

$$\rho(x_0, n_0) := \langle \Phi_{x_0}(t_0 + T, t_0)n_0, n_t \rangle.$$

- The *normal repulsion ratio*:

$$\nu(x_0, n_0) := \frac{\rho(x_0, n_0)}{r(x_0)}.$$

Lagrange coherent structures- Haller's work

A material surface M is *normally repelling over I* , if there exist constants $a, b > 0$ such that for all points $x_0 \in M(t_0)$ and unit normal $n_0 \in N_{x_0}M(t_0)$, we have

$$\rho(x_0, n_0) \geq e^{aT}, \quad \nu(x_0, n_0) \geq e^{bT}.$$

Similarly, we call $M(t)$ *normally attracting over I* if it is normally repelling over I in backward time. $M(t)$ is called *hyperbolic over I* if it is normally repelling or normally attracting over I .

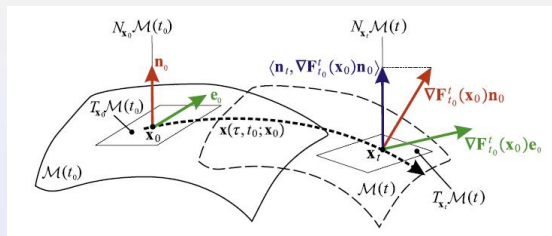


Figure: Lagrange coherent structures

Lagrange coherent structures- Haller's work

M is called repelling weak LCS over I if its normal repulsion rate admits stationary values along $M(t_0)$ among all locally C^1 -close material surfaces. M is called repelling LCS over I if its normal repulsion rate admits a pointwise nondegenerate maximum along $M(t_0)$ among all locally C^1 -close material surfaces.

$$\rho(x_0, n_0) = \frac{1}{\|[\Phi_{x_0}(t_0 + T, t_0)^{-1}]^T n_0\|}.$$

Necessary and sufficient conditions for repelling LCS: Haller (2011)

- (i), $\lambda_1(T, t_0, x_0) > \lambda_2(T, t_0, x_0)$;
- (ii), $\xi_1(T, t_0, x_0) \perp T_{x_0} M(t_0)$;
- (iii), $\langle \nabla \lambda_1(T, t_0, x_0), \xi_1(T, t_0, x_0) \rangle = 0$;
- (iv), $\langle \nabla^2 \lambda_1(T, t_0, x_0) \xi_1(T, t_0, x_0), \xi_1(T, t_0, x_0) \rangle < 0$.

FTLE method for detecting LCSs

Example: double gyre flow

$$\begin{aligned}x &= -\pi 0.1 \sin(\pi f(t, x)) \cos(\pi y) \\y &= \pi 0.1 \cos(\pi f(t, x)) \sin(\pi y) \\f(t, z) &= 0.25 \sin\left(\frac{\pi}{5}t\right)z^2 + [1 - 0.5 \sin\left(\frac{\pi}{5}t\right)]z.\end{aligned}$$

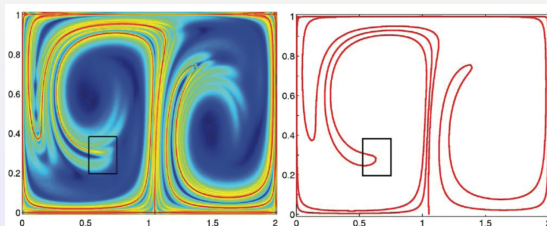


Figure: Double gyre flow - FTLE field

FTLE field sometimes can not detect stable manifolds

- Example 1 - Linear system: $\lambda_1(T, t_0, \cdot) \equiv \lambda$.

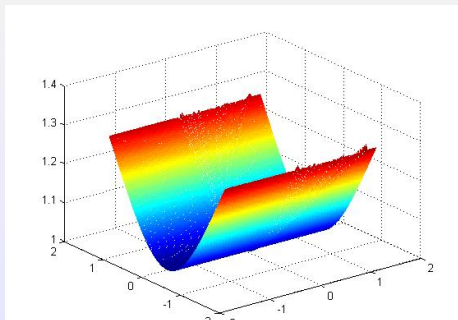
$$\dot{x}_1 = x_1 - x_2$$

$$\dot{x}_2 = -x_2.$$

- Example 2: $M^s(0) = \{(x_1, x_2) : x_2 = \frac{1}{3}x_1^2\}$, $\lambda_1(T, t_0, x_0) = \lambda_1(x_{10})$.

$$\dot{x}_1 = -x_1$$

$$\dot{x}_2 = -x_1^2 + x_2.$$



FT entropy: Froyland and Gehler (2012)

Perron-Frobenius operator: $\mathcal{P}_s^t g(x) = g(\varphi(t, s)^{-1} x)$;

ϵ -smoothing operator: $\mathcal{A}_\epsilon g(x) = \frac{1}{\mu(B(x, \epsilon))} \int_{B(x, \epsilon)} g d\mu = \int_X g f_{B(x, \epsilon)} d\mu$,

for $g \in L^1(X)$, where $f_A(\cdot) := \frac{1}{\mu(A)} 1_A(\cdot)$. Using the *differential entropy* of g defined by $H(g) = - \int_X g \log g d\mu$, the finite time entropy at x_0 at time fiber t_0 is defined by

$$FTE(I_0, x_0) := \lim_{\epsilon \rightarrow 0} \frac{1}{T} [H(\mathcal{A}_\epsilon \mathcal{P}_{t_0}^{t_0+T} f_{B(x_0, \epsilon)}) - H(f_{B(x_0, \epsilon)})].$$

Under the assumption that μ is invariant under φ , we have

$$\begin{aligned} \mathcal{P}_s^t f_{B(x_0, \epsilon)}(\cdot) &= f_{\varphi(t, s)B(x_0, \epsilon)}(\cdot) \\ \mathcal{A}_\epsilon \mathcal{P}_s^t f_{B(x_0, \epsilon)}(\cdot) &= \int_X f_{\varphi(t, s)B(x_0, \epsilon)} f_{B(x, \epsilon)} d\mu = \frac{\mu(\varphi(t, s)B(x_0, \epsilon) \cap B(x, \epsilon))}{\mu(B(x_0, \epsilon))\mu(B(x, \epsilon))}. \end{aligned}$$

FT entropy w.r.t finite partition: Froyland and Gehle (2012)

$(X, \mathcal{F}, \mu)$ - measurable space, $X \subset \mathbb{R}^n$.

$\xi = \{A_1, \dots, A_k\}$ - finite partition of X .

FTE for a set $A \in \xi$

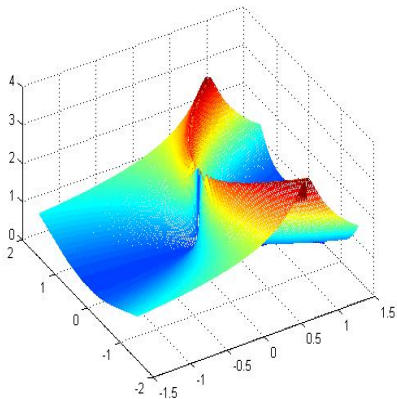
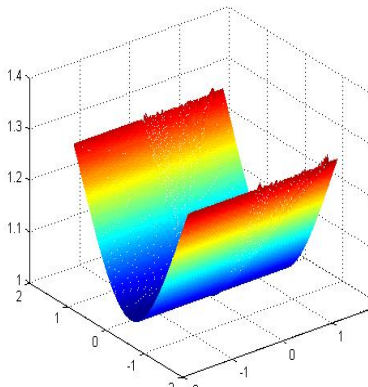
$$FTE(I_0, A) := - \sum_{B \in \xi} \frac{\mu(A \cap \varphi(t_0 + T, t_0)^{-1} B)}{\mu(A)} \log \frac{\mu(A \cap \varphi(t_0 + T, t_0)^{-1} B)}{\mu(A)}. \quad (0.2)$$

FTE field versus FTLE field

Parabol stable manifold: $M^s(0) = \{(x_1, x_2) : x_2 = \frac{1}{3}x_1^2\}$, $\lambda_1(T, t_0, x_0) = \lambda_1(x_{10})$.

$$\dot{x}_1 = -x_1$$

$$\dot{x}_2 = -x_1^2 + x_2.$$



Classical metric entropy

$(X, \mathcal{F}, \mu)$ - measurable space, $X \subset \mathbb{R}^n$.

$T : X \rightarrow X$ - measure preserving transformation.

$\xi = \{A_1, \dots, A_k\}$ - finite partition of X .

$\xi \vee \eta := \{A \cap B : A \in \xi, B \in \eta\}$.

$T^{-i}\xi = \{T^{-i}A_1, \dots, T^{-i}A_k\}$.

Entropy of finite partition: $H_\mu(X, \xi) = -\sum_{i=1}^k \mu(A_i) \log \mu(A_i)$.

Entropy of T : $h_\mu(X, T, \xi) := \lim_{n \rightarrow \infty} \frac{1}{n} H_\mu(X, \bigvee_{i=0}^{n-1} T^{-i}\xi)$.

Metric entropy - Bowen's work

- consider X a compact metric space, and for any $x \in X$ and $n \in \mathcal{N}$, $\epsilon > 0$, define

$$B^\lambda(x, n, \epsilon) := \{y \in X : d(T^i x, T^i y) < \epsilon, 0 \leq i \leq n\}.$$

- Bowen (1971): in case (T, μ) is ergodic,

$$h_\mu(X, T) = \lim_{\epsilon \rightarrow 0} \left\{ \limsup_{n \rightarrow \infty} -\frac{1}{n} \log \mu(B(x, n, \epsilon)) \right\}$$

Metric entropy - Thieullien's work

- consider X a compact metric space, and for any $\lambda \in \mathbb{R}$, $x \in X$ and $n \in \mathcal{N}$, $\epsilon > 0$, define

$$B^\lambda(x, n, \epsilon) := \{y \in X : d(T^i x, T^i y) < \epsilon e^{\lambda i}, 0 \leq i \leq n\}.$$

- Thieullien (1990, 1992): in case T is C^2 diffeomorphism preserving the Lebesgue measure μ . Then

$$\exists h_\mu(\lambda, T, x) := \lim_{\epsilon \rightarrow 0} \lim_{n \rightarrow \infty} -\frac{1}{n} \log \mu(B^\lambda(x, n, \epsilon))$$

- Let $T : M \rightarrow M$ be a C^2 diffeomorphism. Then for almost every $x \in M$

$$h_\mu(\lambda, T, x) = \begin{cases} \sum_{i=1}^n (\lambda_i(x) - \lambda)^+ & \text{if } 0 \leq \lambda \leq -\lambda_n(x), \\ n\lambda & \text{if } \lambda \geq -\lambda_n(x). \end{cases}$$

FT escape rate and FT metric entropy

- For each $\lambda \in \mathbb{R}$ and for a fixed time fiber $s \in J \subset I$, a *fiber metric* $d_{\lambda,s}^{\varphi,J}(\cdot, \cdot) : X \times X \rightarrow \mathbb{R}_+$ by

$$d_{\lambda,s}^{\varphi,J}(y, x) := \sup_{t \in J} \|\varphi(t, s)y - \varphi(t, s)x\| e^{-\lambda(t-s)}.$$

- A λ - ϵ *stable set* of point x_0 at time fiber s is defined by

$$B_{\lambda,s}^{\varphi,J}(x_0, \epsilon) := \{y \in X : d_{\lambda,s}^{\varphi,J}(y, x_0) \leq \epsilon\} \subset B(x_0, \epsilon) := \{y \in X : \|y - x_0\| \leq \epsilon\}.$$

Escape rate and FTME

- The *finite time escape rate* of radius ϵ at point x_0 at time s

$$h_{\lambda,s}^{\varphi,J}(x_0, \epsilon) := -\frac{1}{T} \log \mu(B_{\lambda,s}^{\varphi,J}(x_0, \epsilon) | B(x_0, \epsilon)) = -\frac{1}{T} \log \frac{\mu(B_{\lambda,s}^{\varphi,J}(x_0, \epsilon))}{\mu(B(x_0, \epsilon))}$$

- The λ -*finite time metric entropy* at point x_0 at time s

$$h_{\lambda,s}^{\varphi,J}(x_0) := \lim_{\epsilon \rightarrow 0} h_{\lambda,s}^{\varphi,J}(x_0, \epsilon).$$

Theorem: Properties of FTME

- If $\Phi(t, s) : X \rightarrow X$ is linear for all $t, s \in J$, then $h_{\lambda, s}^{\Phi, J}(x_0)$ is independent of x_0 and is denoted by $h_{\lambda, s}^{\Phi, J}$.
- In general, for each $x_0 \in X$ and $s \in J$: $h_{\lambda, s}^{\varphi, J}(x_0) = h_{\lambda, s}^{\Phi_{x_0}, J}$.
- Boundedness:

$$\frac{n}{T} \log \left(\sup_{t \in J} e^{\lambda(t-s)} \|\Phi_{x_0}(t, s)^{-1}\| \right) \leq h_{\lambda, s}^{\varphi, J}(x_0) \leq \frac{n}{T} \log \left(\inf_{t \in J} \frac{e^{\lambda(t-s)}}{\|\Phi_{x_0}(t, s)\|} \right).$$

- $h_{\lambda, s}^{\varphi, J}(x_0)$ is decreasing in λ and increasing in J .
- Change under the flow

$$h_{\lambda, t}^{\varphi, J}(\varphi(t, s)x_0) = h_{\lambda, s}^{\varphi, J}(x_0) + \frac{n\lambda(t-s)}{T} - \frac{\log |\det \Phi_{x_0}(t, s)|}{T}.$$

- Almost invariance under the change of norm $\|\cdot\|_{\Gamma} := \|\Gamma \cdot\|$ in X , $\Gamma \in \mathbb{R}^{n \times n}$ - positive definite:

$$|h_{\lambda, s}^{\varphi, J, \Gamma}(x_0) - h_{\lambda, s}^{\varphi, J}(x_0)| \leq \frac{n \log \|\Gamma\| + n \log \|\Gamma^{-1}\|}{T}.$$

Pesin's entropy formula

Let X be a compact Riemannian manifold and T is a C^1 diffeomorphism that preserves a smooth Borel probability measure μ . Then

$$h_\mu(T) = \int_X \Sigma(x) d\mu(x) = \int_X \sum \lambda_i^+(x) \dim E_i(x) d\mu(x),$$

where $\lambda^+ = \max\{\lambda, 0\}$, λ_i are the distinct Lyapunov exponents of (T, μ) corresponding to linear subspaces E_i .

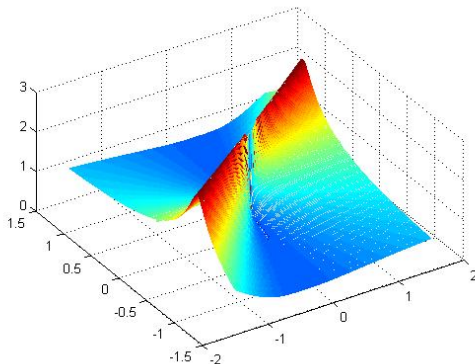
Duc-Siegmund (2014): Pesin's FTME formula

$$|h_{\lambda, t_0}^{\varphi, l_0}(x_0) - \sum_{i=1}^n (\lambda_i(T, t_0, x_0) - \lambda)^+| \leq \frac{n \log 2 - \frac{n}{2} \log \pi + \log \Gamma(\frac{n}{2} + 1)}{T}.$$

Simulation: linear systems

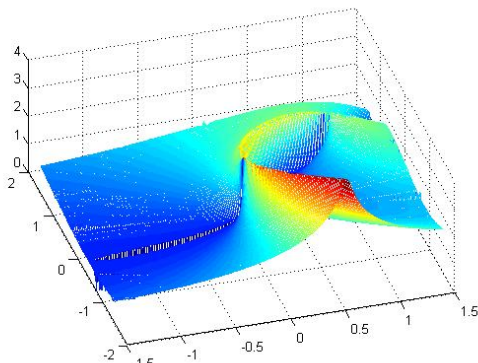
$$\dot{X}_1 = X_1 - X_2$$

$$\dot{X}_2 = -X_2.$$



Simulation: parabol systems

$$\begin{aligned}\dot{x}_1 &= -x_1 + x_2 \\ \dot{x}_2 &= -x_1^2 + x_2.\end{aligned}$$



- Autonomous differential equation

$$\dot{x} = f(x), \quad f \in C^1(X, X), \quad X \subset \mathbb{R}^n.$$

$\varphi(t, t_0, x_0) = \varphi(t - t_0, x_0)$ is a C^1 -function of $x_0 \in X$ and $\Phi_{x_0}(t, t_0) = \Phi_{x_0}(t - t_0) \in C^0(X, \mathbb{R}^n)$.

- $f(x_0)$ is invariant under $\Phi_{x_0}(t)$, i.e.

$$\Phi_{x_0}(t)f(x_0) = f(\varphi(t, x_0)), \quad \forall t \in \mathbb{R}, \forall x_0 \in \mathbb{R}^n.$$

- The *stretching rate*: for any $x_0, f(x_0) \neq 0$,

$$\begin{aligned} \lambda(x_0) &:= \frac{1}{T} \log \frac{\|\Phi_{x_0}(t_0 + T, t_0)f(x_0)\|}{\|f(x_0)\|} \\ &= \frac{1}{T} \log \frac{\|f(\varphi(t_0 + T, t_0, x_0))\|}{\|f(x_0)\|}. \end{aligned}$$

- Duc-Siegmund (2014): weighted FTME field: $H(x_0) := h_{\lambda(x_0), t_0}^{\varphi, l_0}(x_0)$ detects stable and unstable manifolds of hyperbolic autonomous differential systems.

Equip X with a finite time partition ξ , which consists of a finite collection of disjoint compact sets that cover X . In practice ξ has a nice structure such as the regular grid of boxes, triangles, or hexagons. We define

$$r(\xi) := \max\{\text{diam} A : A \in \xi\} = \max\left\{\sup_{x,y \in A} \|x - y\| : A \in \xi\right\}$$

to be the *size of the partition* ξ . For each $x \in X$, denote $\xi(x)$ the element of ξ that contains x .

- The *FTME kernel* w.r.t. the finite partition ξ at (t_0, x_0) is defined by

$$k_{t_0}^{\phi, J}(x_0, \xi, \mu) := -\frac{1}{|J|} \log \frac{\mu(B_{t_0}^{\phi}(x_0, \xi))}{\mu(\xi(x_0))}, \quad (0.3)$$

where

$$\begin{aligned} B_{t_0}^{\phi}(x_0, \xi) &:= \bigcap_{t \in J} \phi(t, t_0)^{-1} \xi(\phi(t, t_0)x_0) \\ &= \{y \in X(t_0) : \phi(t, t_0)y \in \xi(\phi(t, t_0)x_0) \quad \forall t \in J\} \subset \xi(x_0) \end{aligned}$$

is the set of points in X that never leaves the tube $\{t \times \xi(\phi(t, t_0)x_0) : t \in J\}$ under the FTNDS ϕ .

- The *FTME* w.r.t. the partition ξ of the FTNDS ϕ is defined by

$$h_{t_0}^{\phi, J}(X, \xi, \mu) := \frac{1}{|J|} [H_{\mu}(X, \bigvee_{t \in J} \phi(t, t_0)^{-1} \xi) - H_{\mu}(X, \xi)]. \quad (0.4)$$

- The *FTME* w.r.t. the finite partition ξ at (t_0, x_0) is defined by

$$h_{t_0}^{\phi, J}(x_0, \xi, \mu) := h_{t_0}^{\phi, J}(\xi, \mu(\cdot | \xi(x_0))). \quad (0.5)$$

- For $J = I_0 = \{t_0, t_0 + T\}$ and $\varphi = \varphi(t_0 + T, t_0)$ then

$$h_{t_0}^{\phi, J}(x_0, \xi, \mu) := - \sum_{B \in \xi} \frac{\mu(\xi(x_0) \cap \varphi^{-1}B)}{\mu(\xi(x_0))} \log \frac{\mu(\xi(x_0) \cap \varphi^{-1}B)}{\mu(\xi(x_0))}.$$

- For any measurable set M , the *FTME* w.r.t. the finite partition ξ at (t_0, M) is defined by

$$h_{t_0}^{\phi, J}(M, \xi, \mu) := h_{t_0}^{\phi, J}(\xi, \mu(\cdot | M)). \quad (0.6)$$

Theorem

The following equalities hold

$$(i) \quad h_{t_0}^{\phi,J}(X, \xi, \mu) = E[k_{t_0}^{\phi,J}(\cdot, \xi, \mu)] = \int_X k_{t_0}^{\phi,J}(x, \xi, \mu) d\mu(x); \quad (0.7)$$

$$(ii) \quad h_{t_0}^{\phi,J}(x_0, \xi, \mu) = \frac{1}{\mu(\xi(x_0))} \int_{\xi(x_0)} k_{t_0}^{\phi,J}(x, \xi, \mu) d\mu(x); \quad (0.8)$$

$$(iii) \quad h_{t_0}^{\phi,J}(X, \xi, \mu) = E[h_{t_0}^{\phi,J}(\cdot, \xi, \mu)] = \int_X h_{t_0}^{\phi,J}(x, \xi, \mu) d\mu(x). \quad (0.9)$$

Theorem (Duc-Siegmund 2015)

Assume that μ is the Lebesgue measure and ξ is a finite partition of the regular grid. Then the following inequality holds

$$\bar{h}_{t_0}^{\phi,J}(x_0, \mu) := \limsup_{r(\xi) \rightarrow 0} h_{t_0}^{\phi,J}(x_0, \xi, \mu) \geq h_{t_0}^{\phi,J}(x_0, \mu) - \frac{\sqrt{2}}{|J|}$$

FTME field w.r.t. finite partition

The Duffing system is given by the autonomous differential equations

$$\begin{aligned}\dot{x} &= y \\ \dot{y} &= (4 - x^2)x - y,\end{aligned}\tag{0.10}$$

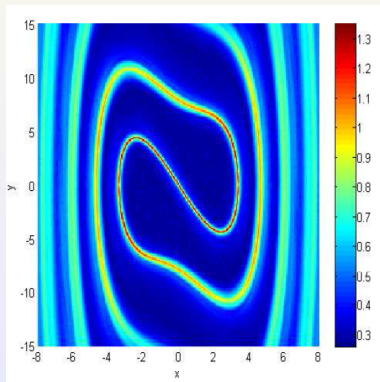


Figure: $J = \{0, 4\}$

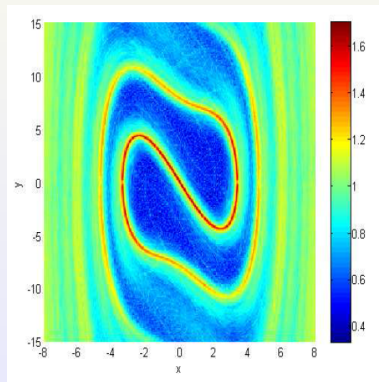
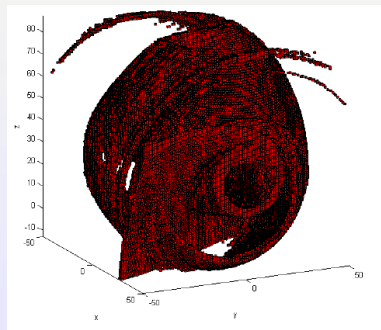
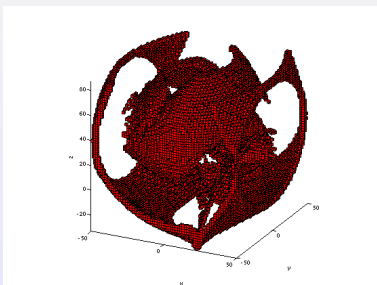


Figure: $J = \{0, 1, 2, 3, 4\}$

The Lorenz flow is given by the autonomous differential equations

$$\begin{aligned}\dot{x} &= \alpha(y - x) \\ \dot{y} &= \rho x - y - xz \\ \dot{z} &= -\beta z + xz,\end{aligned}\tag{0.11}$$

with classical parameters $\alpha = 10, \beta = \frac{8}{3}, \rho = 28$ is well known to possess a chaotic attractor A .



Finite time entropy

- Relation between FTE w.r.t. finite partition and FTLE?
- Relation between FTME and FTE w.r.t finite partition?
- Numerical simulation shows that regions with high FTE detect hyperbolic structures (attracting/repelling material lines/surfaces), while regions with low FTE detect elliptic coherent structures. How can we prove it theoretically?